My research centers on language models (LMs), and methodologically, I am interested in logical reasoning grounded in interactions with humans. I am continually inspired by the adaptability of LMs in addressing increasingly sophisticated problems; however, extending these capabilities to correspond with complex human goals remains challenging due to the inherent difficulty in modeling interactive human behavior and in maintaining logical consistency throughout the reasoning process.

Over the past two years as a master's student, I have been deeply engaged in efforts to operationalize user instructions, enabling LMs to perform effectively on domain-specific tasks. Recently, I have been enthusiastic about exploring reinforcement learning (RL) approaches to foster adaptive interactions between users and models that are grounded in symbolic representations (*e.g.,* code, mathematical expressions). Ultimately, I aim to contribute to the development of scalable LMs capable of self-evolving through continual interaction at the intersection of natural language understanding and symbolic reasoning. Below, I detail my research during my master's program and the future directions I will pursue.

1 Grounding Retrieval-Augmented Generation in Instruction

My first research direction during the master's program focused on aligning LM output with instructions from a user-centric perspective. LMs demonstrate proficient interaction based on their intrinsic, broadly trained knowledge, while their accuracy can be significantly improved when augmented with external sources of information (*e.g.,* Wikipedia, search engines). I have been fascinated by retrieval-augmented generation, not merely for its ability to integrate external information, but for its potential to emulate human processes of searching for, selecting, and verifying information under uncertainty. Therefore, I have become interested in grounding the augmented knowledge that guides output generation directly in the instruction's underlying semantics, ensuring response fidelity to the user's overall goal.

Instruction alignment. As LMs advance, their capabilities are increasingly sought across diverse domains, driving rapid progress in retrieval-augmented methods that effectively incorporate domain-specific knowledge to enhance model reliability. However, I observed that this approach in specific domains (*e.g.,* medical, legal) often yields hallucinations caused by retrieval noise and a lack of domain-expert validation, thereby undermining response reliability. From this observation, I discovered that restructuring the augmented prompt to align strictly with the instruction, without relying on a noisy corpus, could be a promising path forward. In [1], we demonstrated how this approach can guide models toward contextually aligned responses and maintain trust in the augmented prompt, even with noisy domain-specific data.

Personally, working toward my first international publication provided firsthand training in the scholarly process. Our work experienced a few submissions, and although I felt discouraged at first, the process taught me how to critically reflect and systematically strengthen our research. At the same time, I also came to recognize that improving augmented generation in practice requires models to do more than pattern-match retrieved evidence. This work suggests that traditional supervised learning is limited in its ability to capture underlying optimal performance, eventually constraining its ability to generalize beyond the distribution of human-provided examples. LMs learned only the functions or surface-level "what" that are spoon-fed by the data, while lacking the procedural "how" required to accomplish tasks, thereby causing fundamental challenges to scalability. One potential direction I am excited about is to develop methods that extend beyond training on final outputs alone, enabling models to utilize their own thought trajectories. Allowing models to evaluate and refine both their reasoning paths and the external evidence could help overcome the inherent limitations of shallow pattern-copying.

2 Bridging Reasoning and Reinforcement Learning

Another intriguing direction in my research is the use of formal, or more specifically symbolic languages, to elicit elaborate problem-solving in LMs. The recent success of using symbolic languages for reasoning has been a promising advancement, as it shows that explicit representations of knowledge can guide models toward greater interpretability (*e.g.,* chain-of-thought, structured planning). I find this progress particularly exciting, as it suggests the potential for enhancing explainability in training through explicitly modeled decision-making processes. I am keen to explore how LMs with symbolically represented knowledge can enable interpretable reasoning chains that allow systematic observation of model behavior. My interests also extend to the intersection of reasoning and human feedback training, where I aim to explore interactive processes that leverage symbolic representations for real-time verification and refinement of model decisions.

Formal language and verification. One area I have explored is how to operationalize formal languages within LMs to enhance their capabilities as programming assistants. In [2], we demonstrated that LMs could successfully function by internalizing the domain-specific structures of factory control logic through program symbolization into Python-

based representations, and to the best of our knowledge, this was the first attempt in this domain. Given that such assistants are intended for use by factory programmers, we reformulated the task as a preference optimization problem grounded in their programming patterns. This work inspired me to explore intriguing possibilities where LMs can be extended to support cross-domain reasoning.

A limitation of this work is that, although it was structured using formal languages, it lacked integrated logical processes such as software error verification and debugging, which occasionally resulted in hallucinations or calculation errors. Therefore, as a future direction during my doctoral studies, it would be interesting to develop methods that integrate LMs with formal verification and symbolic reasoning to enhance their reliability and logical consistency. However, confining LMs to static verification is insufficient to address symbolic open-ended reasoning (*e.g.,* coding, mathematics) challenges. Reasoning trajectories inherently require an iterative, trial-and-error search over vast combinatorial spaces, and while formal verification can evaluate the correctness of a given trajectory, it fundamentally lacks the capacity to autonomously guide the generation of more effective or productive reasoning paths. Thus, by bridging RL with formal languages, I would like to develop LMs capable of self-evolving through open-ended exploration in their environments, while preserving logical rigor and interpretability.

Toward self-evolving AI. Additionally, a key question in my research is whether learning from human feedback can yield automated reasoning capabilities aligned with humans, and how such capabilities could develop into self-evolving LMs that autonomously generate, evaluate, and refine their own reasoning trajectories. In [3, 4], we introduced an RL approach that continuously interacts with users and iteratively refines code. By incorporating a reward mechanism designed for multi-turn interactions, this method not only demonstrated improved code generation but also exhibited the capacity to refine code in response to modification instructions solely through the model's intrinsic abilities.

To function as a practical decision-making assistant, a model should adequately reflect user needs; however, this work lacked fine-grained human interaction. For example, the instruction was typically framed to ask the model to "fix the code" without providing explicit feedback or indicating what the underlying error was. Therefore, during my PhD, I intend to investigate how these mechanisms might be adapted to facilitate tighter alignment between models and humans. Specifically, I am interested in exploring whether learning from human feedback-driven approaches (*i.e.,* RLHF) can lead to practical reasoning capabilities that go beyond human-level reasoning, and ultimately, how to establish RL environments that enable LM assistants to not only engage in continuous learning and exploration but also cultivate the self-improving behaviors needed for open-ended reasoning. Relatedly, as increasingly complex tasks requiring open-ended reasoning paths demand more sophisticated reward modeling, investigating how reward-based methods can serve as a foundation for consistent LM response presents an exciting research avenue.

3 Future Plans at UMD

I am excited about the opportunity to join the UMD PhD program, where I can further mature as a researcher. I am looking forward to working with professors Tianyi Zhou, Furong Huang, and Heng Huang. Prof. Tianyi Zhou's extensive work on reasoning sets the stage for exploring symbolic languages and self-evolving LMs. I strongly believe that I can further my work and interest in symbolic reasoning under their guidance, especially in human-in-the-loop settings. With Prof. Furong Huang, given our shared interest in the intersection of decision-making and feedback learning, I would like to work on understanding how alignment mechanisms (*e.g.,* RLHF, test-time alignment, reward modeling) can shape long-horizon behaviors in dynamic environments and, ultimately, pave the way for agentic LMs grounded in symbolic languages. With Prof. Heng Huang, I would like to explore how learning across diverse modalities (*e.g.,* natural language, code) can support greater logical coherence in LMs, and how fine-grained reward modeling can further promote consistent patterns in sophisticated, open-ended tasks.

My long-term goal is to become a principal investigator in an academic research lab, a role that perfectly aligns with my passion for collaboration, knowledge-sharing, and addressing critical issues through research. My academic and personal growth will enable me to become a well-rounded leader, equipped with the insight and capability to make meaningful contributions to both academia and society. I highly value the collaborative, inclusive environment of the UMD PhD program, which I believe will help me broaden my research perspective and conduct highly impactful work.

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* indicates co-first author

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